

Yufan Zhuang

✉ y5zhuang@ucsd.edu
☎ +1 (646)-301-4804
🌐 <https://evanzhuang.github.io>
🔗 [linkedin.com/in/yufan-zhuang](https://www.linkedin.com/in/yufan-zhuang)

PROFILE

Research in LLM post-training (continuous/agentic learning, long context understanding). First author of **Test-time Recursive Thinking**, which enabled open models to reach 100% on AIME-25 without external feedback. Multiple first author papers at top venues (NeurIPS, ICLR, ACL, EMNLP, TMLR, FSE, ...), with US/Global patents.

EDUCATION

University of California San Diego, La Jolla, CA Sep 2021 - Mar 2026
PhD in Computer Science, Department of Computer Science & Engineering, Advisor: Prof. Jingbo Shang
Research Interests: Natural Language Processing, Large Language Models' Reasoning and Agentic Learning
Columbia University, New York, NY Aug 2018 - Dec 2019
MS in Data Science, Data Science Institute, GPA: 3.96 / 4.00
Hong Kong Polytechnic University, Kowloon, HK Sep 2014 - May 2018
BSc (Hons) with First Class Honors in Applied Mathematics, Minor in Computer Science, GPA: 4.00 / 4.00

EMPLOYMENT

Research Scientist, Google Cloud AI Research 2026 - Present
· Researching on the frontier of AI & Agents
Research Intern, Microsoft AI, Post-training 2025 - 2026
· Researching on infinite test-time scaling for LLM, shows open models can reach 100% on AIME-25/24 without external feedback
· Designing RL framework for strategic recursive thinking, for coding and math
Machine Learning Intern, Apple Siri 2025 Summer
· Lead research on agentic evaluation systems for next-generation personal mobile assistants
· Expected technical report on simulating realistic human users via finetuning GPT-4o
Research Scientist Intern, AMD GenAI 2024 - 2025
· Pioneered agentic reasoning systems for long context understanding, resulting in +14.7% on HELMET benchmark.
· First author paper AgenticLU published at ACL 2025 main conference (top NLP venue)
Research Scientist Intern, Meta Reality Labs 2024 Summer
· Pretrained efficient VLMs for high-definition OCR, reducing inference latency by 50% while improving general QA performance
· Architected and pretrained Viper vision language models, viper-mamba-7b and viper-jamba-52b
Research Intern, Microsoft Research, Deep Learning Group 2023 Summer
· Pretrained MetaTree, a transformer tabular model over 1M+ datasets for decision tree generation
· First author paper published at TMLR, 100+ stars on github, 10K+ total downloads on Huggingface
Research Engineer, IBM T. J. Watson Research Center 2020 - 2021
· Explored ways for neural methods to understand the logic structure of source code for better robustness and interpretability
· Published 8 papers and 4 patents (2 global patents, 2 US patents)

PUBLICATIONS

LLM REASONING & AGENTIC LEARNING

Yufan Zhuang, Liyuan Liu, Dinghuai Zhang, Chandan Singh, Yelong Shen, Jingbo Shang, and Jianfeng Gao. "Knowledge Flow: Scaling Reasoning Beyond the Context Limit." *Notion Blog*, 2025.
Yufan Zhuang, Liyuan Liu, Chandan Singh, Jingbo Shang, and Jianfeng Gao. "Text Generation Beyond Discrete Token Sampling." *NeurIPS*, 2025.
Yufan Zhuang, Chandan Singh, Liyuan Liu, Jingbo Shang, and Jianfeng Gao. "Vector-ICL: In-context Learning with Continuous Vector Representations." *ICLR*, 2025.
Yufan Zhuang, Xiaodong Yu, Jialian Wu, Ximeng Sun, Ze Wang, Jiang Liu, Yusheng Su, Jingbo Shang, Zicheng Liu, Emad Barsoum. "Self-Taught Agentic Long Context Understanding." *ACL*, 2025.
Yufan Zhuang, Liyuan Liu, Chandan Singh, Jingbo Shang, and Jianfeng Gao. "Learning a Decision Tree Algorithm with Transformers." *Transactions on Machine Learning Research*, 2024.
Feng Yao*, **Yufan Zhuang***, Zihao Sun, Sunan Xu, Animesh Kumar, Jingbo Shang. "Data Contamination Can Cross Language

Barriers." *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024. (*equal contribution)

EFFICIENT TRANSFORMER ATTENTION

Yufan Zhuang, Pierce Chuang, Yichao Lu, Abhay Harpale, Vikas Bhardwaj, and Jingbo Shang. "Viper: Open Mamba-based Vision-Language Models." <https://huggingface.co/ViperVLM>, 2024.

Yufan Zhuang, Zihan Wang, Fangbo Tao, Jingbo Shang. "WavSpA: Wavelet Space Attention for Boosting Transformers' Long Sequence Learning Ability." *NeurIPS UniReps: the First Workshop on Unifying Representations in Neural Models*, 2023.

AI FOR SOFTWARE ENGINEERING

Sahil Suneja, **Yufan Zhuang**, Yunhui Zheng, Jim Laredo, Alessandro Morari, Udayan Khurana. "Incorporating Signal Awareness in Source Code Modeling: An Application to Vulnerability Detection." *ACM Transactions on Software Engineering and Methodology*, Vol. 32, No. 6, Article 145, pp 1–40, 2023.

Sahil Suneja, **Yufan Zhuang**, Yunhui Zheng, Jim Laredo, Alessandro Morari, Udayan Khurana. "Code Vulnerability Detection via Signal-Aware Learning." *IEEE 8th European Symposium on Security and Privacy (EuroS&P)*, 2023.

Yufan Zhuang, Sahil Suneja, Veronika Thost, Giacomo Domeniconi, Alessandro Morari, Jim Laredo. "Software Vulnerability Detection via Deep Learning over Disaggregated Code Graph Representation." *arXiv:2109.03341*, 2021.

Sahil Suneja, Yunhui Zheng, **Yufan Zhuang**, Alessandro Morari, Jim Laredo. "Towards Reliable AI for Source Code Understanding." *ACM Symposium on Cloud Computing (SOCC) Vision Track*, 2021.

Sahil Suneja*, Yunhui Zheng*, **Yufan Zhuang***. "Probing Model Signal-Awareness via Prediction-Preserving Input Minimization." *ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)*, 2021. (*equal contribution)

Luca Buratti, Saurabh Pujar, Mihaela Bornea, Scott McCarley, Yunhui Zheng, Gaetano Rossiello, Alessandro Morari, Jim Laredo, Veronika Thost, **Yufan Zhuang**, Giacomo Domeniconi. "Exploring Software Naturalness through Neural Language Models." *arXiv:2006.12641*, 2020.

Sahil Suneja, Yunhui Zheng, **Yufan Zhuang**, Jim Laredo, Alessandro Morari. "Learning to map source code to software vulnerability using code-as-a-graph." *arXiv:2006.08614*, 2019.

AI FOR SOCIOLOGY

Qiang Fu, **Yufan Zhuang**, Yushu Zhu, Xin Guo. "Sleeping Lion or Sick Man? Combining Computational Approaches to Deciphering Heterogeneous Images of Chinese in North America, 1978-2019." *Annals of the American Association of Geographers*, 2022.

Qiang Fu, **Yufan Zhuang**, Jiaxin Gu, Yushu Zhu, Xin Guo. "Agreeing to Disagree: Choosing among Topic-Modeling Methods." *Big Data Research*, 2020.

Qiang Fu, **Yufan Zhuang**, Jiaxin Gu, Yushu Zhu, Huihui Qin, Xin Guo. "Search for K: Assessing Five Topic-Modeling Approaches to 120,000 Canadian Articles." *BPOD workshop at IEEE International Conference on Big Data*, pp. 3640–3647, 2019.

PATENT

Sahil Suneja, **Yufan Zhuang**, Yunhui Zheng, Alessandro Morari, Jim Alain Laredo, "Artificial intelligence model learning introspection", US/WO Patent, No. US20230130781A1, 2023

Sahil Suneja, **Yufan Zhuang**, Yunhui Zheng, Alessandro Morari, Jim Alain Laredo, "Training data augmentation via program simplification", US/TW/WO Patent, No. US20230113733A1, 2023

Sahil Suneja, **Yufan Zhuang**, Yunhui Zheng, Alessandro Morari, Jim Alain Laredo, "Complexity based artificial intelligence model training", US/CN/JP Patent, No. US20230115723A1, 2023

Sahil Suneja, **Yufan Zhuang**, Yunhui Zheng, Alessandro Morari, Jim Alain Laredo, "Probing Model Signal Awareness", US Patent, No. US20220358400A1, 2023

PROFESSIONAL ACTIVITIES

Reviewer: OOPSLA'20, TSE'21, AAAI'21, OOPSLA'21, NeurIPS'23 (UniReps), WWW'23, NeurIPS'24 (XAI), ICML'24, WWW'24, ICLR'25, TMLR, NeuRIPS'25, ICLR'26, ICML'26

Teaching Assistant: CSE 250A (F'22, F'23, F'24), CSE 251A (S'23), CSE 151A (W'25), CSE 257 (W'23, W'26), CSE 150B (S'25), CSE 101 (F'26), DSC 148 (W'24), DSC 258R (S'24)

SELECTED ACCOMPLISHMENTS AND AWARDS

Jacobs School of Engineering Fellowship	2021
Department of Applied Mathematics Scholarship for Hall Residents	2017/18
The Hong Kong Polytechnic University (Eastern Canada) Association Scholarship	2017/18
The Hong Kong Polytechnic University Scholarship	2016/17
HKSAR Government Scholarship - Reaching Out Award	2015/16
Dean's List	2014/15, 2016/17, 2017/18
Second Prize in National Olympiad in Informatics	2011